

MANY NEW EDGE-TRANSITIVE CONVEX POLYTOPES

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ABSTRACT. We construct many new edge-transitive convex polytopes from exceptional Steiner matroids, signed matroid bases, the deep layer of the binary Golay code, the complement of an extended Hamming code, and nonsingular binary moment matrices. The constructions give eight individual examples and two infinite families; all are in fact arc-transitive. We determine their complete edge graphs, full Euclidean symmetry groups, dimensions, vertex numbers, degrees, metric parameters and second graph eigenvalues. We also determine the least reflection index admitting a vertex-transitive generated subgroup. Midpoint obstructions and alternating principal minors give the edge descriptions; exact finite computations certify the exceptional group actions. Several distinct examples share parameters; one has the same dimension, vertex number, degree, radius and second eigenvalue as the uniform D_8 root polytope. After two explicit identifications, the examples are distinct and extend the known Wythoffian cases. Their full symmetry groups confirm that they do not arise from finite reflection groups. None is uniform: each contains a square-pyramid face or a pyramid over an octahedron, whose graph is not vertex-transitive.

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For more information, see: martinwintermath.github.io/ai-math.html

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1. INTRODUCTION

A convex polytope is *edge-transitive* if its Euclidean symmetry group acts transitively on its edges. This condition imposes restrictions beyond equal edge lengths. In dimension at least four, edge transitivity implies vertex transitivity [5]. A vertex- and edge-transitive polytope is determined up to similarity by its edge graph, and every graph automorphism is induced by a Euclidean symmetry [12, Theorem 4.5].

The known constructions include orbits of finite reflection groups. We call a polytope *Wythoffian* if its vertices form such an orbit in its intrinsic Euclidean space. The classification of finite reflection groups describes the available reflection groups [6]. Edge-transitive Wythoffian examples include regular polytopes, hypersimplices, signed hypersimplices, demicubes and certain rectifications, bitruncations and runcinations of regular polytopes [12, Section 4.4]. These examples motivated the conjecture that every edge-transitive polytope in dimension at least four is Wythoffian and uniform [12, Conjecture 4.19].

We construct eight individual edge-transitive polytopes and two infinite families. All are *arc-transitive*: the symmetry group acts transitively on ordered pairs of adjacent vertices. We determine their edge graphs, full symmetry groups and parameters. The full groups have no vertex-transitive subgroup generated by ordinary reflections, which distinguishes these examples from the Wythoffian constructions. None of the examples is uniform. Equal-scale Cartesian powers of known examples give further examples; we restrict attention to the constructions below.

1.1. The constructions. We first give the three constructions that yield individual examples, then the two infinite families. Their vertex sets are as follows.

- **Exceptional matroid base polytopes.** We take indicator vectors of bases of the matroids defined by the Steiner systems $S(3, 4, 8)$, $S(5, 6, 12)$ and $S(5, 8, 24)$. These give three polytopes of dimensions 7, 11, 23 (Section 2).
- **Signed matroid lifts.** We assign independent signs to the nonzero coordinates of the basis indicators. The three matroids and the dual of the rank-six large-Witt matroid give four polytopes of dimensions 8, 12, 24, 24 (Section 3).

- **The Golay deep-layer polytope.** We take the binary words at Hamming distance four from the extended binary Golay code. This gives one polytope of dimension twenty-four (Section 4).
- **Hamming-complement polytopes.** We take the even binary words outside the extended Hamming code. The resulting polytope H_m has dimension 2^m for every $m \geq 3$ (Section 5).
- **Nonsingular binary-moment polytopes.** We select even binary words by nonsingularity of the matrix of their degree-zero, degree-one and degree-two binary moments. The polytope P_m has dimension 2^m for odd $m \geq 5$. The member P_3 repeats the first matroid base polytope (Section 6).

For each construction we determine all convex edges before deducing arc transitivity; transitivity on shortest vertex pairs alone would not exclude additional edges. Basis exchange and equal-midpoint identities exclude the remaining pairs. The design, code or symplectic actions then prove arc transitivity. The edge directions restrict the full symmetry group to coordinate permutations or signed coordinate permutations. Each construction section also proves nonuniformity. The last section identifies repeated polytopes and compares examples with equal parameters (Section 7). Exact finite computations certify the exceptional group actions; the programs and their outputs are described in Appendix A.

1.2. Conventions and common parameters. All polytopes are convex and considered in their intrinsic affine hull. We center them at their vertex centroid. Their full Euclidean symmetry groups are then finite orthogonal groups, denoted by $\text{Aut}(P)$. An arc is an ordered pair of adjacent vertices. We write $d(P) = \dim P$, $n(P) = f_0(P)$ and $\delta(P)$ for the dimension, vertex number and common graph degree, respectively. For a vertex- and edge-transitive polytope, $R(P)$ is its circumradius and $\ell(P)$ its edge length. The graph Laplacian is $L = \delta I - A$, and λ_2 is its smallest positive eigenvalue. We omit the argument P when it is unambiguous. In particular, $f_1 = n\delta/2$.

The *index* of an orthogonal involution is the dimension of its negative eigenspace. An involution of index k is a *k-reflection*; an *ordinary reflection* has index one. The *transitive reflection index* $k_{\min}(P)$ is the least positive k for which a vertex-transitive subgroup of $\text{Aut}(P)$ is generated by involutions all of index exactly k , and is ∞ if no such subgroup exists. Thus P is Wythoffian precisely when $k_{\min}(P) = 1$. This minimum does not concern individual involutions. For example, in the two infinite families $k_{\min} = 2^{m-2}$, although some members have smaller-index involutions: those involutions preserve each syndrome or moment fiber and cannot generate a vertex-transitive subgroup.

Spectral reconstruction [12, Theorem 4.5] gives the common parameter calculation. The full symmetry group of a vertex- and edge-transitive polytope is irreducible, equals the automorphism group of its edge graph, and its coordinate realization uses the entire second eigenspace. Consequently,

$$(1.1) \quad \sum_{w \sim v} (p_w - p_v) = -\lambda_2 p_v, \quad \lambda_2 = \frac{\delta \ell^2}{2R^2}.$$

The second identity follows by taking the scalar product with p_v . The multiplicity of λ_2 is d .

The tables use the same headings and order: $\dim, f_0, \deg, f_1, R^2, \ell^2, \lambda_2, k_{\min}, |\text{Aut}(P)|$, followed by *cs*, which records whether the polytope is centrally symmetric. The base-polytope table also gives the facet number f_{d-1} . We retain the edge-length column even though all listed realizations have $\ell^2 = 2$. Family tables give initial members; the corresponding theorems state the general parameters. Coordinate equalities defining faces are written before centering.

1.3. Uniformity and the face obstructions. A polytope is *uniform* if it is vertex-transitive and its facets are recursively uniform, with segments and regular polygons as the initial cases. In particular, every face of a uniform polytope is vertex-transitive. For the polytopes considered here every 2-face is a regular polygon: it lies on the intersection of a plane with the circumsphere, and its edges have equal lengths. Uniformity therefore requires additional restrictions on higher-dimensional faces.

Vertex transitivity, equal edge lengths and the group actions underlying our constructions make uniformity a natural possibility to examine. New uniform polytopes would have been a surprising consequence of these constructions. We therefore determine whether the examples are uniform.

To disprove uniformity, we exhibit a face whose edge graph has vertices of different degrees. We use only two face types. A square pyramid occurs in the base polytope of every connected nonuniform matroid (Theorem 2.5); this also excludes uniformity for the signed lifts and moment polytopes. A 4-dimensional pyramid over an octahedron occurs in the Golay and Hamming-complement polytopes (Propositions 4.4 and 5.6). Each face is constructed in the section where it is first used, and subsequent sections refer to the same obstruction. These examples leave open whether uniform edge-transitive polytopes exist outside the Wythoffian class.

2. EXCEPTIONAL MATROID BASE POLYTOPES

Our first three examples are base polytopes of exceptional Steiner matroids. They first arose in Chayim Lowen's answer to the MathOverflow question on matroids transitive on minimal basis exchanges [7]. We determine the edge graphs, parameters and full symmetry groups, then prove nonuniformity by a general result for matroid base polytopes. The same matroids define the signed lifts in Section 3.

2.1. Three Steiner matroids. Let $\mathcal{D}$ be a Steiner system $S(r-1, h, s)$ on a ground set E : every $(r-1)$ -subset is in a unique block of size h . Declare an r -set to be a basis if it is contained in no block. This defines a matroid $M(\mathcal{D})$. Indeed, for bases B, B' and $e \in B \setminus B'$, put $T = B - e$. If every replacement from $B' \setminus B$ were forbidden, all those elements would lie in the unique block through T . The whole of B' would then be in that block, contrary to its being a basis.

We use the affine planes of $\text{AG}(3, 2)$, forming $S(3, 4, 8)$, and the Witt designs $S(5, 6, 12)$ and $S(5, 8, 24)$. Write their matroids as $\mathcal{M}_8, \mathcal{M}_{12}, \mathcal{M}_{24}$, with ranks 4, 6, 6, respectively. The subscript records the ground-set size. Their permutation automorphism groups are $\text{AGL}(3, 2)$ and the Mathieu groups M_{12} and M_{24} , respectively [2, 9]. For $D \subseteq E$, write $\mathbf{1}_D$ for its indicator vector and $\mathbf{1} = \mathbf{1}_E$. For a matroid M of rank r on E , define

$$B(M) = \text{conv}\{\mathbf{1}_B \mid B \in \mathcal{B}(M)\}.$$

We use the centered realization $\mathbf{1}_B - (r/s)\mathbf{1}$.

Theorem 2.1. *The polytopes $B(\mathcal{M}_8), B(\mathcal{M}_{12}), B(\mathcal{M}_{24})$ are arc-transitive, with the parameters in Table 1. Their full symmetry groups are, respectively,*

$$\text{AGL}(3, 2) \times C_2, \quad M_{12} \times C_2, \quad M_{24}.$$

Table 1. Parameters of the exceptional base polytopes.

Polytope	dim	f_0	deg	f_1	R^2	ℓ^2	λ_2	$k_{\min}$	$ \text{Aut}(P) $	cs	f_{d-1}
$B(\mathcal{M}_8)$	7	56	12	336	2	2	6	2	2688	yes	30
$B(\mathcal{M}_{12})$	11	792	30	11880	3	2	10	4	190080	yes	156
$B(\mathcal{M}_{24})$	23	113344	90	5100480	9/2	2	20	8	244823040	no	807

2.2. Edge description. We first justify the edge description used in the parameter table. The following two elementary facts will also be used in the binary and signed constructions.

Lemma 2.2. *Every word in a subset of $\{0, 1\}^E$ is a vertex of its convex hull.*

Proof. For a word x , the functional $z \mapsto \sum_{i \in E} (2x_i - 1)z_i$ uniquely maximizes at x in the entire cube. $\square$

Lemma 2.3. *Let p, q, p', q' be vertices of a polytope with $p + q = p' + q'$ and $p', q' \notin [p, q]$. Then $[p, q]$ is not an edge.*

Proof. A functional exposing $[p, q]$ would also maximize at p' and q' , because the two vertex pairs have the same midpoint. $\square$

By [Lemma 2.2](#), every basis indicator is a vertex. The edges are exactly single basis exchanges. For bases B, B' with $|B \setminus B'| \geq 2$, symmetric basis exchange gives $e \in B \setminus B'$ and $f \in B' \setminus B$ such that $B - e + f$ and $B' - f + e$ are bases. Their indicators have the same sum as $\mathbf{1}_B, \mathbf{1}_{B'}$, so [Lemma 2.3](#) excludes an edge. For a single exchange, fixing all agreeing coordinates leaves exactly the two basis indicators and hence exposes their edge.

The three matroids are connected. Every pair extends to an $(r - 1)$ -set, and hence lies in an r -circuit inside its block. Therefore $d(B(M)) = s - 1$, and every direction $e_i - e_j$ occurs [[4](#), Proposition 2.4].

If b is the number of blocks, then

$$b = \frac{\binom{s}{r-1}}{\binom{h}{r-1}}, \quad n = \binom{s}{r} - b \binom{h}{r}, \quad \delta = r(s - h - 1), \quad R^2 = \frac{r(s - r)}{s}.$$

Different blocks cannot contain the same r -set. For a basis B , each of its r sets $B - e$ forbids $h - r + 1$ of the $s - r$ outside replacements, giving the degree. [Equation \(1.1\)](#) gives $\lambda_2 = \delta/R^2$.

An arc is specified by a common $(r - 1)$ -set and an ordered pair of distinct extension points outside its block. The affine group is transitive on three-sets; after identifying the complementary affine plane with $\mathbb{F}_2^2$, its three-set stabilizer induces $\text{AGL}(2, 2)$ there. This is 2-transitive on the four extension points.

For the small Witt design, a five-set stabilizer has order 120 and is 2-transitive on the six allowed extensions. For the large Witt design, the five-set stabilizer is

$$2^4 \rtimes ((S_5 \times S_3) \cap A_8),$$

of order 5760, acting 2-transitively on the sixteen allowed extensions [[2](#), Table I]. In both cases the full group is transitive on five-sets. These actions prove arc transitivity. For an exact verification of the ordered-pair actions, `verify_seed_polytopes.py` constructs the designs and checks the ordered extension-pair orbits, of sizes 30 and 240, respectively.

We compute the facets using matroid flats. The proper flats are the sets of size at most $r - 2$ and the blocks. A flat inequality defines a facet when restriction and contraction are connected [4, Proposition 2.6]. Sets of size $2, \dots, r - 2$ have disconnected free restrictions; each block gives $U_{r-1,h} \oplus U_{1,s-h}$ and hence a facet. The faces $x_i = 1$ are facets because M/i is connected: circuits in blocks containing i connect every pair of remaining elements after contraction. The faces $x_i = 0$ are facets because $M \setminus i$ is connected: the number of blocks through a pair exceeds that through a triple, so each pair of remaining elements lies in a circuit avoiding i . These give exactly $b + 2s$ facets.

2.3. Full symmetry and reflection index. The edge directions are the roots $A_{s-1} = \{e_i - e_j \mid i \neq j\} \subseteq \mathbf{1}^\perp$. Their orthogonal automorphisms are coordinate permutations and their negatives [6].

Lemma 2.4. *An involutory permutation of s coordinates with t transpositions has index t on $\mathbf{1}^\perp$. Its negative has index $s - 1 - t$ there.*

Proof. Each transposition contributes one negative eigenvalue. The vector $\mathbf{1}$ is fixed, so restriction to $\mathbf{1}^\perp$ does not change that count. Negation interchanges the positive and negative eigenspaces. $\square$

All A_{s-1} directions occur, so every full symmetry is a coordinate permutation or its negative. The first two basis families are self-complementary; their negatives give the central C_2 . For the large Witt example, the two centered coordinate values are $3/4$ and $-1/4$, so a negative permutation cannot preserve the vertices. The permutation parts are exactly the design automorphisms. This proves the full symmetry group assertions in [Theorem 2.1](#).

By [Lemma 2.4](#), the relevant indices are the transposition counts. The least transposition counts of nonidentity involutions in the three permutation groups are 2, 4, 8, respectively. For the Mathieu groups the involution types are $2^4 1^4, 2^6$ in degree twelve and $2^8 1^8, 2^{12}$ in degree twenty-four [10, 11]. Negative permutations in the two central extensions have indices at least three and five, respectively, and do not lower these minima.

Affine transvections of index two generate $\text{AGL}(3, 2)$. The minimum-index involution classes generate the simple Mathieu groups, and these groups are already vertex-transitive. Thus $k_{\min} = 2, 4, 8$. The script `verify_parameters.py` additionally enumerates every element of the degree-twelve group, checks its involution indices, and checks generation by the 495 involutions of index four. For degree twenty-four it constructs the index-eight conjugacy class of size 11 385; generation follows from simplicity. In particular, none of the base polytopes has an ordinary reflection.

2.4. Nonuniformity. The three matroids are connected and nonuniform: their defining blocks contain dependent sets of size equal to the rank. The following theorem therefore applies. We state it for arbitrary matroids because it will also be used for the signed lifts and moment polytopes.

Theorem 2.5. *A standard matroid base polytope is uniform if and only if its matroid is a direct sum of uniform matroids, with loops and coloops allowed. In fact, every connected nonuniform matroid has a square-pyramid face in its base polytope.*

Proof. Let M be connected and nonuniform of rank r . Choose a circuit C with $|C| \leq r$, take $e \in C$, and extend $C - e$ to a basis B . The fundamental bipartite graph between B and $E \setminus B$ has an edge $\{b, f\}$ exactly

when $B - b + f$ is a basis. It is connected because M is connected, but not complete: the fundamental circuit of e is a proper subset of $B + e$. A connected noncomplete bipartite graph contains an induced path on four vertices, obtained from a shortest path between a nonadjacent pair in opposite parts.

Contract the other basis elements and delete the other nonbasis elements. The resulting rank-two four-element matroid has fundamental graph that three-edge path, hence exactly one parallel pair. Its five bases give

$$\text{conv}\{e_i + e_j \mid 1 \leq i < j \leq 4, \{i, j\} \neq \{1, 2\}\}.$$

This is a square pyramid: the apex has degree four and the other vertices degree three. The contraction and deletion are realized by fixing coordinates to one and zero, respectively, so this is a face of $B(M)$.

For a general matroid, the base polytope is the Cartesian product of component base polytopes. A nonuniform component retains its obstructing face after fixing vertices in the other factors. Conversely, uniform components give hypersimplices, with their common standard edge length $\sqrt{2}$. Their faces are products of hypersimplices, recursively uniform. Loops and coloops contribute point factors. $\square$

Each of $B(\mathcal{M}_8), B(\mathcal{M}_{12}), B(\mathcal{M}_{24})$ therefore has a square-pyramid face and is not uniform. For $B(\mathcal{M}_8)$, an explicit face is obtained by imposing $x_0 = x_1 = 1$ and $x_5 = x_7 = 0$ on binary labels $0, \dots, 7$. The remaining coordinates $2, 3, 4, 6$ give five basis indicators.

3. SIGNED MATROID LIFTS

We now assign independent signs to the nonzero coordinates of each basis vector. This gives four further polytopes, including distinct lifts of a matroid and its dual. We prove that their edges correspond to single basis exchanges with consistent signs, then determine the symmetry groups and exhibit a nonuniform face.

For a connected rank- r matroid M on s elements, with $0 < r < s$, define its *signed lift* by

$$Q(M) = \text{conv}\{y \in \{0, \pm 1\}^s \mid \text{supp}(y) \in \mathcal{B}(M)\}.$$

Assigning signs removes the constraint $\sum_i y_i = r$; every vertex instead has squared norm r . We use $\mathcal{M}_8, \mathcal{M}_{12}, \mathcal{M}_{24}$ and, in the last case, its rank-eighteen dual $\mathcal{M}_{24}^*$.

3.1. Edge description.

Lemma 3.1. *In a finite vertex set on a sphere centered at the origin, every pair at minimum distance is an exposed edge of the convex hull.*

Proof. Let p, q be a closest pair, with $\|p\| = \|q\| = R$ and $\|p - q\| = \ell$. For every other vertex r , $\|r - p\|, \|r - q\| \geq \ell$, so

$$\langle p + q, r \rangle \leq 2R^2 - \ell^2 < 2R^2 - \ell^2/2 = \langle p + q, p \rangle = \langle p + q, q \rangle.$$

Thus $p + q$ exposes precisely the segment $[p, q]$. $\square$

Lemma 3.2. *The edges of $Q(M)$ are exactly the pairs whose supports differ by one basis exchange and whose signs agree on the common support. If the base graph is regular of degree δ_B , then*

$$n(Q(M)) = 2^r n(B(M)), \quad \delta(Q(M)) = 2\delta_B.$$

Proof. Each signed basis vector is exposed by its own scalar product. The stated pairs have minimum distance $\sqrt{2}$ and are edges by [Lemma 3.1](#).

It remains to exclude other pairs. Opposite signs on a common coordinate give the obstruction in [Lemma 2.3](#) by reversing that sign at both endpoints, unless this is their only difference. For a lone sign change, exchange its coordinate with one outside the basis and give the entering coordinate opposite signs. Such an exchange exists because there is no coloop. If the signs agree on the common support, two or more support exchanges reduce by symmetric basis exchange: transfer the leaving and entering signed entries between the endpoints. This again preserves the midpoint. Each single exchange has two entering signs, giving the degree formula. $\square$

Theorem 3.3. *The four signed lifts in [Table 2](#) are arc-transitive. Their full groups are $(C_2)^s \rtimes \text{Aut}(M)$, with $\text{Aut}(M)$ the matroid permutation group. All are centrally symmetric.*

Table 2. Parameters of the signed matroid lifts. The squared circumradius equals the rank of the underlying matroid.

Polytope	dim	f_0	deg	f_1	R^2	ℓ^2	λ_2	$k_{\min}$	$ \text{Aut}(P) $	cs
$Q(\mathcal{M}_8)$	8	896	24	10 752	4	2	6	2	$2^8 \cdot 1\,344$	yes
$Q(\mathcal{M}_{12})$	12	50 688	60	1 520 640	6	2	10	4	$2^{12} \cdot 95\,040$	yes
$Q(\mathcal{M}_{24})$	24	7 254 016	180	652 861 440	6	2	30	8	$2^{24} \cdot 244\,823\,040$	yes
$Q(\mathcal{M}_{24}^*)$	24	29 712 449 536	180	2 674 120 458 240	18	2	10	8	$2^{24} \cdot 244\,823\,040$	yes

3.2. Full symmetry and reflection index. The signed edge directions are the roots $D_s = \{\pm e_i \pm e_j \mid i \neq j\} \subseteq \mathbb{R}^s$. The following restriction on symmetries will also apply to the code and moment polytopes.

Lemma 3.4. *For $s \geq 5$, the orthogonal automorphisms of D_s are exactly the signed coordinate permutations. Consequently, a polytope whose complete edge-vector set is D_s has only signed coordinate permutations as Euclidean symmetries.*

Proof. The first assertion is the orthogonal automorphism description of the root system D_s [6]. A Euclidean symmetry permutes the edge vectors, which proves the second assertion. $\square$

Lemma 3.5. *A signed permutation involution has one negative eigenvalue for each transposed coordinate pair and one for each negative fixed coordinate. Its ordinary reflections are single-coordinate sign changes and signed transpositions.*

Proof. On a transposed pair the involution condition forces the two signs to agree, and the two eigenvalues are 1, -1 . A fixed coordinate contributes the sign assigned to it. $\square$

Proof of [Theorem 3.3](#). The base-exchange action is transitive on ordered basis exchanges. Independent signs then match either endpoint and its entering sign. Thus [Lemma 3.2](#) gives arc transitivity. Every vertex has squared norm r , and negation is a symmetry, so [Equation \(1.1\)](#) gives $\lambda_2 = 2\delta_B/r$.

Connectedness of M gives every exchange direction $e_i - e_j$, and independent signs give all D_s directions. These span $\mathbb{R}^s$ and, by [Lemma 3.4](#), restrict the full symmetries to signed permutations. Their supports force the permutation part to preserve M , proving the full symmetry group assertion.

Let $k = 2, 4, 8$ be the minimum involution index of its permutation group. Every smaller-index signed involution is diagonal by [Lemma 3.5](#), and preserves each support. Such involutions cannot generate a vertex-transitive subgroup. For the upper bound, the index- k permutation involutions generate the permutation group, while diagonal changes on k coordinates generate all even sign changes. Since k is even and $2 \leq k \leq s - 2$, symmetric differences of k -sets give all coordinate pairs. Even sign changes prescribe arbitrary signs on a support of rank $r < s$ by correcting parity outside it. This gives a vertex-transitive subgroup generated by exact k -reflections. $\square$

Here ordinary reflections do exist: they are the coordinate sign changes. They leave every basis support invariant, and hence cannot give a Wythoffian orbit. The rank-six and rank-eighteen lifts also show that matroid duality behaves differently before and after adding signs.

3.3. Nonuniformity. The nonuniformity proof reduces to the base-polytope theorem. Maximizing $\sum_i y_i$ on $Q(M)$ exposes the face consisting of all-positive basis vectors, which is $B(M)$. For each of the four matroids used here, [Theorem 2.5](#) gives a square-pyramid face of $B(M)$ and hence of $Q(M)$. For the large-Witt dual, connectedness and nonuniformity follow from duality, since duality preserves connectedness and takes uniform matroids to uniform matroids.

More generally, a signed lift of a matroid with a nonuniform connected component has the same obstruction. If all components are uniform, their signed lifts are type- B reflection-group orbits, and the product is again a reflection-group orbit. Thus signed matroid lifts cannot produce a uniform non-Wythoffian polytope.

4. THE DEEP LAYER OF THE BINARY GOLAY CODE

We now pass from matroids to binary codes. For the extended binary Golay code, we take the words farthest from the code. Its distance-four cosets determine partitions into six tetrads, which determine the coordinate graphs. This produces a single example in dimension twenty-four, with the same vertex number and radius as the rank-six signed large-Witt lift.

For a binary word x , $|x|$ denotes Hamming weight, and $\oplus$ denotes binary addition.

Let $C \leq \mathbb{F}_2^{24}$ be the extended binary Golay code, of dimension twelve and minimum weight eight. Define

$$G = \text{conv}\{x - \tfrac{1}{2}\mathbf{1} \mid \text{dist}(x, C) = 4\}, \quad \text{dist}(x, C) = \min_{c \in C} |x \oplus c|.$$

We call a word *deep* if its distance from C is four. The code is even, so every deep word is even. The Golay code has covering radius four. Its five distance layers contain 1, 24, 276, 2024 and 1771 cosets, respectively. The minimum-weight words in a coset are its leaders. Every distance-four coset has six weight-four leaders, whose disjoint supports partition the coordinates into a *sextet* of four-element sets, called tetrads [[2](#), pp. 1040–1041, Tables I–II]. The coordinate group M_{24} is transitive on sextets, and a sextet stabilizer is transitive on ordered pairs from different tetrads. These two actions are checked explicitly by `verify_seed_polytopes.py`.

Theorem 4.1. *The Golay deep-layer polytope G is arc-transitive, with the parameters in [Table 3](#). Its full symmetry group is $C \rtimes M_{24}$.*

Table 3. Parameters of the Golay deep-layer polytope.

Polytope	dim	f_0	deg	f_1	R^2	ℓ^2	λ_2	$k_{\min}$	$ \text{Aut}(P) $	cs
G	24	7 254 016	240	870 481 920	6	2	40	8	$2^{12} \cdot 244 823 040$	yes

4.1. Edge description. A *two-coordinate flip* replaces x by $x \oplus \mathbf{1}_{\{a,b\}}$ with $a \neq b$. A word is called feasible if its centered vector is in the displayed vertex set defining the polytope. The *coordinate graph* at a feasible word x has the coordinate labels as vertices; $\{a, b\}$ is an edge exactly when that two-coordinate flip is feasible. We use these conventions throughout the three binary constructions.

Lemma 4.2. *In the convex hull of a subset of even binary words, every feasible two-coordinate flip is an exposed edge.*

Proof. All selected words are vertices by [Lemma 2.2](#). Fixing the agreeing coordinates of the two words leaves only two even assignments on the remaining pair. $\square$

At a deep word, the feasible two-coordinate flips are exactly the pairs from different tetrads of its sextet. A pair in one leader tetrad leaves distance two from a codeword. Conversely, if a two-coordinate flip leaves distance at most two, its union with the residual error gives a weight-four leader of the original coset. The pair is therefore contained in one tetrad.

Consider two deep words whose difference set D has size greater than two. It cannot be contained in a tetrad of either sextet: its even size would then be four and the toggle would give a codeword. We claim that D contains a pair in different tetrads of both sextets. Otherwise every pair crossing the first partition would be in one part of the second. Comparing each point with a point in a different first part forces all of D into one second part, a contradiction. Flipping the resulting pair at both endpoints gives the same midpoint. By [Lemmas 2.3](#) and [4.2](#), the edges are exactly the feasible two-coordinate flips.

4.2. Full symmetry and reflection index. We first obtain arc transitivity from the code action, then recover the code from the vertex set to determine the full symmetry group. The following lemma will also be used for both infinite families.

Lemma 4.3. *Let $C \leq \mathbb{F}_2^E$ be a binary linear code, and let a coordinate permutation group Π preserve C and a union V of its cosets. Then $C \rtimes \Pi$ acts by Euclidean symmetries on $\text{conv}\{x - \frac{1}{2}\mathbf{1} \mid x \in V\}$. Translations by C are transitive on each coset. If $\pi \in \Pi$ preserves the coset $x + C$, then $y \mapsto \pi y \oplus (x \oplus \pi x)$ fixes x .*

Proof. Translation by c acts on centered coordinates as the diagonal orthogonal map D_c with entries $(-1)^{c_i}$. Coordinate permutations are orthogonal as well. The coset condition gives $x \oplus \pi x \in C$, and substitution verifies that the displayed action fixes x . $\square$

Code translations act transitively within a deep coset by [Lemma 4.3](#); the sextet action moves between deep cosets. The ordered cross-tetrad pair action, with the word stabilizer obtained by [Lemma 4.3](#), gives arc transitivity. Each coordinate has twenty feasible partners, so $\delta = 24 \cdot 20/2 = 240$. The coset count gives $n = 1771 \cdot 4096$. The all-ones word is in C ; central symmetry and [Equation \(1.1\)](#) give the stated metric parameters.

The code is self-dual and has no nonzero word supported on a pair. Hence its projection onto any coordinate pair is onto $\mathbb{F}_2^2$. The 2-transitive coordinate action and this sign freedom give all D_{24} directions, proving full dimension and restricting every symmetry to a signed permutation by [Lemma 3.4](#).

There is one more point to check. Deep words are even, so a full symmetry preserves the even cube and its excluded subset. By covering radius and parity, that subset is precisely

$$C \cup \{x \mid \text{dist}(x, C) = 2\}.$$

In its induced distance-two graph, a codeword has degree $\binom{24}{2} = 276$. A word at distance two has a unique nearest codeword and degree $1 + 2(24 - 2) = 45$. No distance-two neighbor associated with another codeword is possible, since it would give a nonzero codeword of weight at most six. Thus the code is intrinsically recoverable as the degree-276 vertices of this excluded graph.

Writing a symmetry as $x \mapsto \pi x \oplus c$, we conclude $c \in C$ and $\pi(C) = C$, hence $\pi \in M_{24}$. This proves the full symmetry group assertion in [Theorem 4.1](#). By [Lemma 3.5](#), the indices are bounded below by eight: nonidentity permutation involutions have at least eight transpositions, and nonzero diagonal code involutions have weight at least eight. Octad indicators span C , while the index-eight involution class generates the simple group M_{24} . Thus $k_{\min} = 8$.

The distinction between the two excluded degrees is essential. Merely knowing that a symmetry preserves the deepest layer would not by itself identify its action on the code.

4.3. Nonuniformity. We exhibit a pyramid over an octahedron as a coordinate face. The same face will exclude uniformity for the Hamming-complement family.

Proposition 4.4. *The polytope G has a face isometric to*

$$T = \text{conv}(\{e_i + e_j \mid 1 \leq i < j \leq 4\} \cup \{(1, 1, 1, 1)\}).$$

This is a 4-dimensional pyramid over a regular octahedron, so G is not uniform.

Proof. Choose a six-set B contained in no Golay octad; such sets are the bases of $\mathcal{M}_{24}$ from [Section 2](#). Fix two of its coordinates to one and all coordinates outside B to zero. These supporting equalities define a face. The resulting weight-two word is at distance two from the zero codeword and is excluded. Each of the six weight-four words is deep: a codeword at distance less than four would contradict minimum distance eight.

The weight-six word is also deep. The unique octad through five points of B meets B in exactly five, and hence gives distance four. A codeword at distance at most two would have to be an octad containing B , which was excluded. Odd words cannot be deep because the code is even and has covering radius four. Exactly the six weight-four words and the weight-six word remain. After subtracting the two fixed coordinates, this face is isometric to T . Its apex has graph degree six and its six base vertices have degree five. Hence its graph is not vertex-transitive. $\square$

5. THE COMPLEMENT OF THE EXTENDED HAMMING CODE

We next construct two infinite families from binary words. Removing an extended Hamming code from the even cube gives the first family. The syndrome determines both the selected vertices and their feasible coordinate flips, so the edge calculation and the affine symmetry action remain explicit for every family member.

Let $m \geq 3$, put $d = 2^m$, and label coordinates by $a \in \mathbb{F}_2^m$. For a word x , write $\sigma(x) = \sum_a x_a a$. The extended Hamming code is

$$C_m = \{x \in \mathbb{F}_2^d \mid |x| \equiv 0 \pmod{2}, \sigma(x) = 0\}.$$

We select its complement in the even cube:

$$H_m = \text{conv}\{x - \tfrac{1}{2}\mathbf{1} \mid |x| \equiv 0 \pmod{2}, \sigma(x) \neq 0\}.$$

The even cube itself gives the Wythoffian demicube. Removing the code leaves its sphere and edge length unchanged, but has a substantial effect on the symmetry.

Theorem 5.1. *For every $m \geq 3$, H_m is an arc-transitive, centrally symmetric polytope of dimension $d = 2^m$. Its parameters are*

$$n = (d-1)2^{d-m-1}, \quad \delta = \frac{d(d-2)}{2}, \quad f_1 = n \frac{d(d-2)}{4}, \quad R^2 = \frac{d}{4}, \quad \ell^2 = 2, \quad \lambda_2 = 2(d-2), \quad k_{\min} = \frac{d}{4}.$$

Its full group is $C_m \rtimes \text{AGL}(m, 2)$, of order $2^{d-m-1} d \prod_{i=0}^{m-1} (d - 2^i)$.

Table 4. Parameters of the first three Hamming-complement polytopes.

Polytope	dim	f_0	deg	f_1	R^2	ℓ^2	λ_2	$k_{\min}$	$ \text{Aut}(P) $	cs
H_3	8	112	24	1 344	2	2	12	2	21 504	yes
H_4	16	30 720	112	1 720 320	4	2	28	4	660 602 880	yes
H_5	32	2 080 374 784	480	499 289 948 160	8	2	60	8	$2^{26} \cdot 319 979 520$	yes

5.1. Edge description. We use the two-coordinate flips and coordinate graphs defined in [Section 4.1](#). The even-word edge lemma [Lemma 4.2](#) applies to the selected vertices.

Proposition 5.2. *The edges of H_m are exactly the feasible two-coordinate flips. At a word of syndrome $\sigma \neq 0$, its coordinate graph is the complete graph on $\mathbb{F}_2^m$ with the pairs $\{a, a + \sigma\}$ removed.*

Proof. At a word with syndrome $\sigma \neq 0$, a two-coordinate flip $\{a, b\}$ is feasible exactly when $a + b \neq \sigma$. The forbidden pairs form a perfect matching. For two feasible words with difference set D of size greater than two, the forbidden pairs at the endpoints belong to two matchings. Their union has at most $|D|$ pairs, fewer than $\binom{|D|}{2}$. Flipping such a pair at both endpoints gives a second vertex pair with the same midpoint.

[Lemma 2.3](#) excludes the longer segment, and [Lemma 4.2](#) proves the converse. □

5.2. Full symmetry and reflection index. We determine the full symmetry group and its transitive reflection index. We use the code action from [Lemma 4.3](#) and record the Reed–Muller and affine-reflection results needed here and for the moment family.

Let $d = 2^m$, with coordinate labels in $\mathbb{F}_2^m$. The Reed–Muller code $\text{RM}(t, m)$ is the space of evaluation vectors of squarefree polynomials of degree at most t . We use the following properties [\[8\]](#).

Lemma 5.3. For $0 \leq t < m$,

$$\dim \text{RM}(t, m) = \sum_{j=0}^t \binom{m}{j}, \quad \text{RM}(t, m)^\perp = \text{RM}(m - t - 1, m),$$

and its minimum nonzero weight is 2^{m-t} . Indicators of affine j -flats span $\text{RM}(m - j, m)$ for $1 \leq j \leq m$.

Proof. The squarefree monomial evaluations are independent. Monomial orthogonality gives the duality: a monomial of degree less than m sums to zero on $\mathbb{F}_2^m$, and the dimensions give equality. Induction on the last variable gives the minimum-weight formula. Indicators of coordinate j -flats are products of $m - j$ affine coordinate functions. Summing over unused fixed coordinates yields all monomials of smaller degree. Affine substitutions preserve the degree bound. $\square$

In particular, $C_m = \text{RM}(m - 2, m)$. Indeed, the defining parity and syndrome equations say that C_m is orthogonal to the degree-at-most-one evaluations. Affine-plane indicators therefore span C_m .

Lemma 5.4. A nonidentity affine involution of $\mathbb{F}_2^m$ moves at least 2^{m-1} points. Every signed permutation involution with this permutation part has index at least 2^{m-2} .

Proof. The fixed set of a nonidentity affine map is either empty or a proper affine subspace, and hence has at most half the points. Every moved pair contributes one negative eigenvalue by Lemma 3.5. $\square$

For a coordinate set T , write $D_T = D_{\mathbf{1}_T}$.

Lemma 5.5. Let T be a proper positive-dimensional affine flat in $\mathbb{F}_2^m$. There is an affine involution τ preserving T such that τ and $D_T\tau$ both have index $d/4$ in the coordinate action. Moreover, affine involutions of this index generate $\text{AGL}(m, 2)$.

Proof. Choose a nonzero direction b of T , and a nonconstant affine function f equal to one on T , whose linear part annihilates its direction space. Then $\tau(a) = a + f(a)b$ is an involution. It moves precisely the $d/2$ points with $f(a) = 1$, including every point of T , and has $d/4$ transpositions. Adding signs on T creates no negative fixed coordinates, so Lemma 3.5 gives the same index for $D_T\tau$. Their product is D_T .

The maps $a_i \mapsto a_i + a_j$ and $a_i \mapsto a_i + a_j + 1$, for $i \neq j$, have index $d/4$. The first maps generate $\text{GL}(m, 2)$; the product of the two displayed maps is a coordinate translation. Together they generate $\text{AGL}(m, 2)$. $\square$

A nonzero translation instead has index $d/2$. The proper-flat hypothesis permits the nonconstant function f and hence the index $d/4$.

Proof of Theorem 5.1. The syndrome map on even words is onto $\mathbb{F}_2^m$, and each fiber has 2^{d-m-1} words. By Lemma 4.3, code translations are transitive on a fiber, while the affine group moves nonzero syndromes transitively. The stabilizer of a nonzero syndrome is transitive on coordinate pairs whose difference avoids $\{0, \sigma\}$: use label translations and linear maps fixing σ . The coset-preserving actions fix the word after the correction in Lemma 4.3. This proves arc transitivity.

There are $\binom{d}{2} - d/2 = d(d-2)/2$ incident flips. The all-ones word is in C_m , so central symmetry, the radius and Equation (1.1) give the remaining metric parameters. The code projects onto all four binary patterns on a pair of coordinates: affine-linear evaluation functions already do so. Together with affine pair

transitivity this gives all D_d directions and full dimension. [Lemma 3.4](#) therefore restricts the full symmetry group to signed permutations.

Every full symmetry is now a signed permutation, written in binary form as $x \mapsto \pi x \oplus c$. Its shift c is even; hence it preserves the even cube and its excluded subset C_m . Thus $c \in C_m$ and $\pi(C_m) = C_m$. The weight-four code supports are precisely the affine planes. After making π fix zero, preservation of $\{0, a, b, a + b\}$ forces additivity. Therefore $\pi \in \text{AGL}(m, 2)$, giving the full group.

Involutions of index below $d/4$ are diagonal by [Lemma 5.4](#), so preserve syndrome and cannot be vertex-transitive. Affine-plane indicators span C_m by [Lemma 5.3](#). [Lemma 5.5](#) produces their diagonal maps from exact $d/4$ -reflections and generates the affine group by the same index. This proves $k_{\min} = d/4 > 1$. $\square$

5.3. Nonuniformity. We exhibit the same pyramid over an octahedron as in [Proposition 4.4](#).

Proposition 5.6. *Every H_m , $m \geq 3$, has a face isometric to the pyramid T from [Proposition 4.4](#), and is not uniform.*

Proof. Fix all coordinates outside $\{0, e_1, e_2, e_3\}$ to zero. These are supporting equalities for the binary hull. The zero word has syndrome zero and is excluded. All six weight-two words have nonzero syndrome, and the weight-four word has syndrome $e_1 + e_2 + e_3 \neq 0$. The resulting seven-vertex face is isometric to T . By [Proposition 4.4](#), its graph is not vertex-transitive. $\square$

6. NONSINGULAR BINARY-MOMENT POLYTOPES

Our second infinite family replaces the syndrome condition by nonsingularity of a binary moment matrix. We determine its edges by an alternating principal-minor identity and prove arc transitivity using a symplectic group action. We also identify the smallest member, which repeats the affine matroid base polytope.

Let $m \geq 3$ be odd. Index the 2^m coordinates by $a \in \mathbb{F}_2^m$, and set $u_a = (1, a)^\top \in \mathbb{F}_2^{m+1}$. Define

$$A(x) = \sum_{a \in \mathbb{F}_2^m} x_a u_a u_a^\top, \quad P_m = \text{conv}\{x - \tfrac{1}{2}\mathbf{1} \mid |x| \equiv 0 \pmod{2}, \det A(x) \neq 0\}.$$

All matrix operations here are over $\mathbb{F}_2$. The determinant condition gives a vertex set invariant under affine coordinate permutations. We will express feasible coordinate flips using principal submatrices of an alternating matrix.

Theorem 6.1. *The member P_3 is exactly the centered polytope $B(\mathcal{M}_8)$. For every odd $m \geq 5$, P_m is an arc-transitive, centrally symmetric polytope of dimension $d = 2^m$. Put $h = (m + 1)/2$. Its parameters are*

$$(6.1) \quad n = 2^{d-h^2-1} \prod_{j=1}^h (2^{2j-1} - 1).$$

$$\delta = \frac{d(d-2)}{4}, \quad f_1 = n \frac{d(d-2)}{8}, \quad R^2 = \frac{d}{4}, \quad \ell^2 = 2, \quad \lambda_2 = d-2, \quad k_{\min} = \frac{d}{4}.$$

Its full group is $K_m \rtimes \text{AGL}(m, 2)$, where $K_m = \text{RM}(m-3, m)$ has dimension $\kappa = d-1-m-\binom{m}{2}$. The group order is $2^\kappa d \prod_{i=0}^{m-1} (d-2^i)$.

Table 5. Parameters of the first two moment polytopes. P_3 is the affine base polytope $B(\mathcal{M}_8)$.

Polytope	dim	f_0	deg	f_1	R^2	ℓ^2	λ_2	$k_{\min}$	$ \text{Aut}(P) $	cs
P_3	7	56	12	336	2	2	6	2	2688	yes
P_5	32	910	163968	240	109219676160	8	2	30	$8 \cdot 2^{16} \cdot 319979520$	yes

6.1. The moment orbit and its size. By [Lemma 5.3](#), the entries of $A(x)$ record the sums of all squarefree monomials of degrees zero, one and two. These evaluation functions are independent. Thus the moment image has dimension $1 + m + \binom{m}{2}$ and

$$K_m = \ker A = \text{RM}(m-3, m), \quad \dim K_m = \kappa = 2^m - 1 - m - \binom{m}{2}.$$

The affine group acts on moments by congruence, and translations by K_m are transitive in every fiber by [Lemma 4.3](#).

For the moment of an even word, the first row and diagonal are both $(0, \sigma)$. Nonsingularity forces $\sigma \neq 0$. After a linear change making $\sigma = e_1$, write the moment as

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & v^\top \\ 0 & v & C \end{pmatrix},$$

where C is alternating of order $m-1$. Its determinant is $\det C$. An affine translation eliminates v , and all nonsingular alternating forms of this order are congruent. Hence the nonsingular moments of even words form one affine orbit, and $K_m \rtimes \text{AGL}(m, 2)$ is vertex-transitive.

The number of nonsingular alternating matrices of order $2t$ is

$$a_{2t} = \frac{|\text{GL}(2t, 2)|}{|\text{Sp}(2t, 2)|} = 2^{t(t-1)} \prod_{j=1}^t (2^{2j-1} - 1).$$

There are $2^m - 1$ choices of nonzero σ , 2^{m-1} choices of v and a_{m-1} choices of C . Multiplying by the fiber size 2^κ and simplifying gives [\(6.1\)](#). For $m = 5$, the nonsingular moment orbit has 13888 elements and the kernel has 65536 words, giving 910163968 vertices.

6.2. Edge description. We prove an alternating-matrix criterion for the edge description. Its application to P_m follows from the determinant identity below, and it will also be used for the graph-pivot construction.

Lemma 6.2. *Let B be a nonsingular alternating matrix over $\mathbb{F}_2$ of order at least four. For every index i there is $j \neq i$ such that $B_{ij} = 1$ and the complementary principal submatrix $B[\overline{\{i, j\}}]$ is nonsingular.*

Proof. The inverse is alternating as well, and $\sum_j B_{ij}(B^{-1})_{ji} = 1$. Choose a nonzero summand. Jacobi's complementary-minor identity gives

$$\det B[\overline{\{i, j\}}] = \det B \cdot \det B^{-1}[\{i, j\}] = \det B \cdot (B_{ij}^{-1})^2 \neq 0.$$

□

Proposition 6.3. *Let $V \subseteq \mathbb{F}_2^E$ consist of even words. Suppose that for every $x \in V$ there is an alternating matrix Γ_x indexed by E such that, for every $D \subseteq E$, $x \oplus \mathbf{1}_D \in V$ if and only if $\Gamma_x[D]$ is nonsingular. Then the edges of $\text{conv } V$ are exactly the feasible two-coordinate flips.*

Proof. For $x, y = x \oplus \mathbf{1}_D \in V$ with $|D| > 2$, apply [Lemma 6.2](#) to $\Gamma_x[D]$. It gives $i, j \in D$ such that both $\Gamma_x[\{i, j\}]$ and $\Gamma_x[D \setminus \{i, j\}]$ are nonsingular. Thus $x' = x \oplus \mathbf{1}_{\{i, j\}}$ and $y' = x \oplus \mathbf{1}_{D \setminus \{i, j\}}$ belong to V and satisfy $x + y = x' + y'$. [Lemma 2.3](#) excludes $[x, y]$. The converse is [Lemma 4.2](#). $\square$

Fix a feasible word x and put $A = A(x)$. Since $\text{diag}(A) = Ae_0$, we have $w^\top Aw = e_0^\top Aw$ over $\mathbb{F}_2$, where e_0 selects the first coordinate. Substituting $w = A^{-1}u_a$ gives $u_a^\top A^{-1}u_a = 1$. Let U have columns u_a . Then $\Gamma = I + U^\top A^{-1}U$ is alternating, and the determinant lemma gives

$$(6.2) \quad \det A(x \oplus \mathbf{1}_D) = \det A(x) \cdot \det \Gamma[D].$$

[Proposition 6.3](#) now proves the complete edge description: the edges are exactly the feasible two-coordinate flips. In particular, Γ is the adjacency matrix of the coordinate graph at x : its $\{a, b\}$ principal minor is nonsingular exactly when $\Gamma_{ab} = 1$.

At the moment

$$A_0 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \oplus J,$$

with J nondegenerate alternating, label coordinates as $(t, z) \in \mathbb{F}_2 \times \mathbb{F}_2^{m-1}$ and let ω be the form from J^{-1} . The coordinate graph is

$$(6.3) \quad (t, z) \sim (t', z') \iff (t, z) \neq (t', z'), \quad t + t' + \omega(z, z') = 0.$$

Every coordinate has $2^{m-1} - 1$ partners. Its affine automorphisms include

$$(t, z) \mapsto (t + c + \omega(Sz, b), Sz + b), \quad S \in \text{Sp}(m-1, 2), \quad b \in \mathbb{F}_2^{m-1}, \quad c \in \mathbb{F}_2.$$

They are transitive on coordinates; the stabilizer of $(0, 0)$ is transitive on its neighbors $(0, z)$ with $z \neq 0$. These transformations preserve the chosen moment. The correction in [Lemma 4.3](#) gives a polytope vertex stabilizer, proving arc transitivity. Counting edges in this coordinate graph gives $\delta = 2^{m-1}(2^{m-1} - 1)$. For $m \geq 5$, this is $d(d-2)/4$.

6.3. Full symmetry and reflection index. A nonsingular moment needs at least $m+1$ selected columns. Equality holds exactly for affine bases of $\text{AG}(m, 2)$. The all-ones word belongs to K_m , so complements are feasible. When $m = 3$, both a feasible word and its complement need at least four columns; hence every word has weight four. This proves $P_3 = B(\mathcal{M}_8)$ in dimension seven.

For odd $m \geq 5$, minimizing $\sum_a x_a$ exposes the affine-basis polytope. The matroid is connected, so its base polytope has dimension $d-1$, whereas complementary bases have the different weight $d-m-1$ and leave that hyperplane. Thus P_m is full-dimensional and the affine-basis polytope is a facet. The group is 2-transitive on coordinate labels, and the kernel projects onto all four patterns on a pair because it contains affine-linear evaluation functions. Hence all D_d directions occur, and [Lemma 3.4](#) restricts the full group to signed permutations.

It remains to exclude additional signed permutations. Move the image of a chosen vertex back using the known transitive group. A signed permutation fixing that vertex is determined by its permutation part,

which must preserve (6.3). The pairs $\{(0, z), (1, z)\}$ are intrinsically characterized as distinct nonadjacent pairs with no common neighbor. Every other nonadjacent pair has 2^{m-2} common neighbors. After composing with one of the displayed affine symplectic maps, an automorphism fixes $(0, 0)$ and $(1, 0)$, and has the form $(t, z) \mapsto (t + \epsilon(z), \phi(z))$. Adjacency to $(0, 0)$ gives $\epsilon(z) = 0$. Adjacency between the remaining pairs gives $\omega(\phi(z), \phi(w)) = \omega(z, w)$. Nondegeneracy and bijectivity imply $\phi(z + w) = \phi(z) + \phi(w)$, so ϕ is symplectic-linear. All these local actions already lift to the known group. Therefore $\text{Aut}(P_m) = K_m \rtimes \text{AGL}(m, 2)$.

For $m \geq 5$, Lemma 5.4 shows that involutions of index below $d/4$ are diagonal and preserve the moment, so cannot generate a vertex-transitive group. Affine three-flat indicators span K_m by Lemma 5.3. Since these flats are proper for $m \geq 5$, Lemma 5.5 gives the reverse bound and $k_{\min} = d/4$. The radius $R^2 = d/4$ and Equation (1.1) give $\lambda_2 = d - 2$. For $m = 3$ we use the affine-base parameters instead, including $d = 7$ and $k_{\min} = 2$. This completes Theorem 6.1.

For $m \geq 7$, the kernel has weight-eight words and hence individual diagonal 8-reflections. Nevertheless, $k_{\min} = d/4 \geq 32$: every involution of index below $d/4$ is diagonal and fixes the moment. Even the group generated by all such smaller-index involutions therefore fails to be vertex-transitive.

6.4. Nonuniformity. The affine-basis face identified above reduces the proof to Theorem 2.5. For $m \geq 5$ it is a facet; for $m = 3$ it is the whole polytope $B(\mathcal{M}_8)$. The affine-basis matroid is connected because every pair lies in an affine-plane circuit. It is nonuniform because such a circuit has four elements, at most its rank $m + 1$. Thus its base polytope contains a square-pyramid face, and so does P_m . Every moment polytope is therefore nonuniform. The explicit five-vertex face for P_3 was given in Section 2.

6.5. A generalization by alternating models. The principal-minor criterion in Proposition 6.3 applies to binary vertex sets beyond those defined by the moment condition. We therefore ask whether it can yield further edge-transitive polytopes. We define feasible words using the adjacency matrix of a graph and use principal pivots to give a sufficient condition for arc transitivity. The instances discussed below recover known polytopes; we establish no additional example. Whether this criterion yields an example beyond the Wythoffian polytopes and the constructions above remains open.

To formulate the criterion, let H be a simple graph with adjacency matrix M over $\mathbb{F}_2$, and define

$$\mathcal{F}_H = \{D \subseteq V(H) \mid M[D] \text{ is nonsingular}\}, \quad Q_H = \text{conv}\{\mathbf{1}_D \mid D \in \mathcal{F}_H\}.$$

The empty principal submatrix is nonsingular by convention.

For $X \in \mathcal{F}_H$, the principal pivot transform $M * X$ is alternating and satisfies

$$D \in \mathcal{F}_{H * X} \iff X \triangle D \in \mathcal{F}_H.$$

For a block matrix with nonsingular block A on X , the transform is

$$\begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} * X = \begin{pmatrix} A^{-1} & A^{-1}B \\ B^\top A^{-1} & C + B^\top A^{-1}B \end{pmatrix}.$$

The displayed feasibility identity follows from the principal-minor identity for this block formula. Thus Proposition 6.3 gives the complete edge description of Q_H ; at the empty vertex, incident edges correspond exactly to graph edges of H .

If $X = \{i, j\}$ is an edge, we call $H * X$ an *edge pivot* of H . Here $H * X$ is the graph with adjacency matrix $M * X$.

Proposition 6.4. *Suppose every edge pivot of H is isomorphic to H , and $\text{Aut}(H)$ is edge-transitive. Then Q_H is arc-transitive in its intrinsic affine hull.*

Proof. Alternating Gaussian elimination factors every feasible principal pivot into successive pivots on graph edges. A pivot-isomorphism is realized on the binary cube by toggling its pivot coordinates and then permuting coordinates. Composing these isomorphisms moves every feasible word to the empty word. Graph edge transitivity then makes the empty-word stabilizer transitive on its incident polytope edges. $\square$

The condition $H * \{i, j\} \cong H$ for every edge $\{i, j\}$ is called ELC preservation. Recursive constructions and finite tables are available [3, Sections 4–6]. The hypotheses do not imply that the resulting polytope is new. Complete graphs give Wythoffian demicubes, and stars give regular simplices. For the cube graph on eight coordinates it recovers P_3 : translate the feasible moment words by an affine-basis indicator and use (6.2). The coordinate graph (6.3) for $m = 3$ is a cube. Indeed, replacing t by $t + z_1 z_2$ makes it a Cayley graph of $\mathbb{F}_2^3$ with three independent generators.

One graph construction gives no further examples. Replace each vertex of a connected graph by a clique with at least two vertices, and join two such cliques completely exactly when the original vertices are adjacent. An edge within a replacement clique has identical closed neighborhoods at its endpoints. If the expanded graph is edge-transitive, every edge has this property; connectedness then forces the entire graph to be complete. Thus this construction satisfies the criterion only when the expanded graph is complete.

The remaining question concerns edge-transitive graphs satisfying ELC preservation beyond these constructions.

Question 6.5. Does there exist an edge-transitive graph H , isomorphic to each of its edge pivots, for which Q_H is neither Wythoffian nor similar to any of the polytopes constructed here?

Any candidate still requires a full symmetry calculation and a comparison with the constructions already given.

7. IDENTIFICATIONS AND PARAMETER COINCIDENCES

We now compare the constructions, first identifying repeated polytopes and then distinguishing examples with equal parameters. The comparison with the D_8 root polytope shows that dimension, vertex and edge counts, radius and second eigenvalue together need not identify the polytope.

Two identifications must be made before counting the examples. First, $P_3 = B(\mathcal{M}_8)$. Second, complementation gives $B(M) \cong B(M^*)$ by $x \mapsto \mathbf{1} - x$ for every matroid. In particular, the dual large-Witt base polytope is not an additional example. The first two matroids are identically self-dual, so their dual signed lifts repeat the same supports. The rank-six and rank-eighteen large-Witt signed lifts remain distinct.

Other equalities of parameters do not give isomorphic polytopes. The polytopes $Q(\mathcal{M}_{24})$ and G both have dimension twenty-four, 7 254 016 vertices, central symmetry, $R^2 = 6$, $\ell^2 = 2$ and $k_{\min} = 8$. Their matching vertex counts are two identities in the same Golay system:

$$113\,344 \cdot 2^6 = 1771 \cdot 2^{12}.$$

Their degrees, however, are 180 and 240, and their second Laplacian eigenvalues are 30 and 40. They are not even combinatorially isomorphic.

Similarly, $Q(\mathcal{M}_8)$ and H_3 share dimension eight, degree 24 and $k_{\min} = 2$, but have 896 and 112 vertices. The polytope H_3 shares additional parameters with the rectified 8-crosspolytope, the D_8 root polytope

$$R_8 = \text{conv}\{\pm e_i \pm e_j \mid 1 \leq i < j \leq 8\}.$$

It is Wythoffian and uniform: its facets are regular 7-crosspolytopes and hypersimplices $\Delta(8, 2)$ [1, Proposition 18]. Both H_3 and R_8 have

$$d = 8, \quad n = 112, \quad \delta = 24, \quad f_1 = 1344, \quad R^2 = 2, \quad \ell^2 = 2, \quad \lambda_2 = 12,$$

and are centrally symmetric. The identity $R_8 = Q(U_{2,8})$ and the signed-lift edge description show that each vertex of R_8 is adjacent to the 24 roots obtained by replacing one of its two support coordinates, with either sign, by one of the six remaining coordinates.

The graphs nevertheless differ locally. At a vertex of H_3 , the feasible flips are the edges of the complete graph on eight coordinates with a perfect matching removed. Two flips give adjacent neighbors precisely when they share a coordinate. Thus the induced neighbor graph is the line graph of a 6-regular graph and has degree ten. At $e_1 + e_2$ in R_8 , the neighbors fall into two copies of the 6-crosspolytope graph, joined by a matching. Their induced graph has degree eleven. The edge graphs are therefore not isomorphic. Their full symmetry orders are also different. For R_8 , Lemma 3.4 gives the signed coordinate permutation group:

$$|\text{Aut}(H_3)| = 21\,504, \quad |\text{Aut}(R_8)| = 2^8 8! = 10\,321\,920.$$

The program `verify_parameters.py` enumerates both graphs and checks the common counts and the two local degrees.

In dimension seven, $B(\mathcal{M}_8)$, the hypersimplex $\Delta(8, 3)$ and the Wythoffian Gosset polytope 3_{21} all have 56 vertices. Their degrees are 12, 15 and 27, respectively [12, Theorem 4.18]. Thus equal vertex numbers do not imply equal edge graphs.

The Hamming-complement family is obtained from the Wythoffian demicube by deleting its codeword vertices. It lies on the same sphere as the d -demicube, with the same edge length, and differs by removing just a fraction $1/d$ of its vertices. The demicube has 2^{d-1} vertices and degree $\binom{d}{2}$. The Hamming-complement polytope has $(d-1)2^{d-m-1}$ vertices and degree $\binom{d}{2} - d/2$. Its full group nevertheless has no ordinary reflections.

Apart from the two identifications above, dimension, vertex number or degree distinguishes all the displayed examples. For odd $m \geq 5$, the Hamming and moment families have a common dimension $d = 2^m$ and degrees $d(d-2)/2$ and $d(d-2)/4$.

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APPENDIX A. REPRODUCIBLE FINITE VERIFICATION

The preceding proofs use finite design and group actions at several specific points. We make those inputs independently verifiable with exact programs and record their outputs here. The programs also check the parameter tables and finite instances of the edge arguments.

The supplementary material contains four Python programs, a runner and the two ATLAS permutation-generator files M24.g1 and M24.g2. All programs use exact integer arithmetic, binary-matrix elimination and permutation actions. They require Python 3.10 or newer and no third-party packages. The runner is `python3 run_all.py`; assertions must be enabled. The complete runner was executed with Python 3.12.14 and all four programs passed. It writes the output logs, JSON results and a manifest with source-file hashes.

- `verify_binary_inputs.py` reconstructs all 4 096 Golay words and their cosets, checks the covering radius, all 1 771 sextets and the octad Steiner property, and verifies the seven-vertex deep-layer face. It checks all words for H_3 and P_3 , including longer-pair midpoint witnesses and their seven- and five-vertex faces. It enumerates all 32 768 even moment matrices for $m = 5$, finds 13 888 nonsingular matrices, and checks the entire 65 536-word kernel with minimum weight eight.
- `verify_seed_polytopes.py` constructs the large Witt design from the ATLAS generators and obtains the small Witt design by restricting a Golay dodecad stabilizer. It checks the single base orbits of sizes 113 344 and 792 and their degrees 90 and 30; Schreier generators give the required extension-pair orbits of sizes 240 and 30. It also checks all 896 signed affine-base vertices and their nonneighbor witnesses. For the deep layer it checks all 488 796 coset/pair feasibility tests, the sextet orbit and the ordered cross-tetrad pair orbit of size 480. It enumerates all 7 254 016 deep words relative to one vertex and constructs midpoint witnesses for all 7 253 775 nonneighbors. It checks all 30 720 vertices of H_4 and their nonneighbor witnesses. Finally, it reaches every nonsingular $m = 5$ moment in one affine orbit, checks the orbit of incident edges of size 240 and the antipodal pairs in the coordinate graph, and tests all 942 152 difference sets of sizes four and six at a canonical feasible word.
- `verify_structural_properties.py` checks the exact reflection-generator indices and kernel spanning for H_3, H_4, P_5, P_7 . It verifies that exact 2-reflections generate $\text{AGL}(3, 2)$, and checks the principal-minor edge argument on all 1 024 labelled graphs with five vertices.
- `verify_parameters.py` checks the numerical tables, the design inputs for the facet counts, all degree-twelve involutions and generation by the minimum-index class, and an explicit degree-twenty-four index-eight involution with its conjugacy class. Its parameter output also records full group orders and edge numbers for sample members of the infinite families. It also checks the vertex and edge counts and the distinct induced neighbor graphs of H_3 and the uniform D_8 root polytope.

The ATLAS files are parsed as permutation cycles. Neither GAP nor numerical convex-hull software is required. The programs do not enumerate all 910 163 968 vertices of P_5 or the rank-eighteen signed lift; those counts follow from the verified fiber or support counts. The proofs give the edge descriptions, full symmetry groups and reflection indices for every family member.